United States Patent	12410830
Kind Code	B2
Date of Patent	September 09, 2025
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Access cover assembly

Abstract

An access cover assembly for machines, equipment, mechanical assemblies and like apparatus is configured to be mounted or attached to an exterior housing or casing of the apparatus to cover an access port or opening in the housing that is used for gaining access to the interior of the apparatus. The access cover assembly includes a cover plate and a removable, self-locking plug that can be quickly and easily installed and removed from the cover plate. The cover plate has a threaded access opening and the self-locking plug includes a threaded shaft that engages with the threaded access opening when secured to the cover plate. The self-locking plug incorporates a lock ring that inhibits unintended rotation and loosening of the self-locking plug when assembled to the cover plate.

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Appl. No.: **18/115170**

Filed: **February 28, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230287927 A1	Sep. 14, 2023

Related U.S. Application Data

us-provisional-application US 63317733 20220308

Publication Classification

Int. Cl.: **F16B39/284** (20060101); **F16B2/06** (20060101); **F16B2/10** (20060101); **F16M1/00** (20060101)

U.S. Cl.:

CPC **F16B39/284** (20130101); **F16B2/06** (20130101); **F16B2/10** (20130101); **F16M1/00** (20130101);

Field of Classification Search

CPC: F16B (39/284); F16B (39/282); F16B (2/06); F16B (2/10); F16M (1/00); F16F (1/38); F01D (21/003); B64F (5/60)

USPC: 411/190; 411/166; 411/326; 411/327; 411/329; 411/951; D16/130

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 63/317,733, filed on Mar. 8, 2022. The entire disclosure of the above application is incorporated herein by reference.

FIELD
(1) The present disclosure relates to a removable cover for machines, equipment, mechanical assemblies and the like that enable access to internal features and/or components of the apparatus.
BACKGROUND
(2) This section provides background information related to the present disclosure which is not necessarily prior art.
(3) Machines, equipment, mechanical assemblies and the like often include panels or covers that are removably fastened or affixed to a housing or casing of the apparatus by use of fasteners. See FIG. **1**, Prior Art. The access covers enable access to the internal workings, features and/or components of the apparatus to enable the apparatus to be inspected, maintained, serviced or repaired. Access covers are known to be used on aircraft and land-based turbine engines.
(4) Conventional access covers are attached to the apparatus by a plurality of fasteners (e.g., threaded fasteners such as mounting bolts) that can be tightened to a specified torque. In some applications, it is important to further ensure that the fasteners attaching the access cover to the apparatus do not loosen during normal operation or use of the apparatus. One known manner in which to lessen a tendency for the attachment bolts to loosen is by securing the mounting bolts together with a wire that counteracts a loosening of the fastener.
(5) Lock wire, also known as safety wire, is often used in situations where mechanical reliability is critical. For example, in aircraft applications, safety wire is used to secure the fasteners attaching the access covers to mechanical assemblies, as well as with other fasteners in the aircraft. This is because the significant vibration and other forces that act on the aircraft and its components can contribute to the fasteners inadvertently becoming loose.
(6) In order to remove an access cover that has been fastened and lockwired to an apparatus, however, such as for inspecting the apparatus, the lock wire needs to be cut off and disconnected from the mounting bolts before the mounting bolts can be loosened and removed. Thereafter, the access cover can be removed to make the needed internal workings of the apparatus available to service personnel. After the inspection or other work is performed, the mounting bolts are used to reattach the access cover in place on the apparatus and the fasteners must be re-lockwired. This

process requires a particular skill set, is very labor intensive and time consuming. Thus, the costs to perform routine maintenance, inspection or other service activities are unnecessarily increased.

(7) Furthermore, in some applications, the maintenance, inspection or service activity needed to be performed on the apparatus does not require as much space or availability as is afforded by the complete removal of the access cover. Nevertheless, conventional access covers must be removed in their entirety in order to gain any access to the internal features or components of the apparatus. Still further, the access cover on some apparatus may be located in a manner such that there is limited space to easily employ or manipulate the tools that are needed to remove the lock wire, unfasten the fasteners and/or visually locate the access cover. These conditions add further difficulties, labor and cost to otherwise routine tasks.

SUMMARY

(8) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(9) The present disclosure provides an access cover assembly for machines, equipment, mechanical assemblies and like apparatus. The access cover assembly is configured to be mounted or attached to an exterior housing or casing of an apparatus to cover an access port or opening in the housing that is used for gaining access to the interior of the apparatus. The access cover assembly includes a cover plate and a removable, self-locking plug that can be quickly and easily installed and removed from the cover plate. The cover plate has an access opening and the self-locking plug that can be secured to the cover plate. The self-locking plug incorporates a lock ring that inhibits unintended rotation and loosening of the self-locking plug when assembled to the cover plate.

(10) In one aspect of the disclosure, an access cover assembly can including a cover plate and a self-locking plug removably attached to the cover plate. The cover plate can include a central portion and a mounting flange positioned about a perimeter of the central portion. The flange can have a plurality of apertures configured to accommodate a plurality of mounting bolts for attaching the access cover assembly to, e.g., a piece of machinery or equipment. The cover plate can also have a first aperture having a first diameter and a second aperture having a second diameter that is greater than the first diameter. The first aperture can be centered on a central axis and extend through the central portion. The first aperture can include a first inner circumferential wall that has an internal screw thread. The second aperture can also be centered on the central axis and be concentric with the first aperture. The second aperture extend into the central portion to a depth that is less than a thickness of the central portion (e.g., like a counter-bore). The second aperture can have a second inner circumferential wall with a plurality of concave-shaped detents equally spaced about a perimeter of the second inner circumferential wall. Adjacent detents can be separated from one another by a substantially planar or substantially cylindrical surface.

(11) The self-locking plug can be disposed in the first aperture and the second aperture along the central axis. The self-locking plug can include a main body and a lock ring attached to the main body. The main body can have a head and a shaft, with the head being configured to receive a tool for rotating the self-locking plug about the central axis. The shaft can include an external screw thread.

(12) The lock ring can include a cylindrical portion attached to the head and a plurality of resilient, flexible beam fingers arranged about a circumference of the main body. Each beam finger can extend from a proximal end at the cylindrical portion to a distal end comprising a fingertip. Each fingertip of the plurality of beam fingers can include a plurality of radially inwardly beveled surfaces and a substantially planar surface between the beveled surfaces, The fingertip can be configured to mechanically interact with the second inner circumferential wall of the second aperture so as to inhibit rotation of the self-locking plug about the central axis.

(13) In another aspect of the disclosure, an access cover assembly can include a cover plate and a removable, threaded, self-locking plug configured to detachably connect to the cover plate. The cover plate can have a detent interface that is co-operable with the self-locking plug. The detent

interface can include a boss projecting from a central portion of the cover plate, the boss having a second diameter and including a second outer circumferential wall with a cylindrical surface. In the second outer circumferential wall a plurality of spaced-apart, concave-shaped detents are included that are equally spaced about a perimeter of the second outer circumferential wall. Adjacent detents can be separated from one another by discrete sections of the cylindrical surface of the outer circumferential wall.

(14) A first aperture having a first diameter and being concentric with the boss can be included in the cover plate. The first aperture can extend along an axis through both the boss and the central portion of the cover plate. The first aperture can also include an inner circumferential wall defining an internal screw thread. The boss can define an interface port for the self-locking plug and the first aperture can define an access opening through the cover plate.

(15) The self-locking plug can include a main body and a lock ring that is attached to the main body. The main body can have a head and a shaft, and the head can have a tool engaging interface and the shaft can have an external screw thread. The lock ring can include an upper section that is attached to the head of the main body and a plurality of resilient, flexible beam fingers that are arranged about a circumference of the upper section. Each beam finger can extend from a cylindrical portion of the upper section of the lock ring to a distal end that includes a plurality of beveled surfaces and is configured to mechanically interact with the detent interface to inhibit rotation of the self-locking plug about the axis.

(16) In still another aspect of the disclosure, an access cover assembly can include a cover plate and a removable, threaded, self-locking plug configured to detachably connect to the cover plate. The cover plate can have a detent interface that is co-operable with the lock ring to inhibit unintended loosening of the plug. The detent interface can have a boss projecting from a central portion of the cover plate and the boss can include a cylindrical outer wall having an outer diameter. The cylindrical outer wall of the boss can define an external screw thread. A first aperture of the cover plate can include a first diameter and extend along an axis through the boss and the central portion of the cover plate. A second aperture in the boss can include a second diameter that is greater than the first diameter and concentric with the cylindrical outer wall of the boss, The second aperture can have a depth that is less than a height of the boss and include a plurality of spaced-apart, concave-shaped detents that are equally spaced about a perimeter of the second aperture. Adjacent detents can be separated from one another by discrete cylindrically shaped sections of the second aperture. The boss can define an interface port for the self-locking plug and the first aperture can define an access opening through the cover plate.

(17) The self-locking plug can include a cylindrical, cap-like main body and a lock ring that is attached to the main body. The main body can have a sleeve including an internal, cylindrically shaped surface defining an internal screw thread.

(18) The lock ring can include an upper section attached to the main body and a plurality of resilient, flexible beam fingers arranged about a circumference of the upper section. Each beam finger can extend from a cylindrical portion of the upper section to a distal end, which distal end include a plurality of beveled surfaces and is configured to mechanically interact with the detent interface to inhibit rotation of the self-locking plug about the axis.

(19) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and

not all possible implementations and are not intended to limit the scope of the present disclosure.
(2) FIG. **1** shows a conventional, prior art, access cover;
(3) FIG. **2** shows an isometric view of an access cover assembly according to the principles of the present disclosure;
(4) FIG. **3** shows a front view of the access cover assembly of FIG. **2**;
(5) FIG. **4** shows a right side view the access cover assembly of FIG. **2**;
(6) FIG. **5** shows an exploded isometric view of the access cover assembly of FIG. **2**;
(7) FIG. **6** shows an exploded right side view of the access cover assembly of FIG. **2**;
(8) FIG. **7** shows a cross-sectional view along the section line **7-7** of the access cover assembly of FIG. **3**;
(9) FIG. **8** shows an isometric view of the cover plate of the access cover assembly of FIG. **2**;
(10) FIG. **9** shows a front view of the cover plate of FIG. **8**;
(11) FIG. **10** shows a back view of the cover plate of FIG. **8**;
(12) FIG. **11** shows a front view of the self-locking plug of the access cover assembly of FIG. **2**;
(13) FIG. **12** shows a top view of the self-locking plug of FIG. **11**;
(14) FIG. **13** shows an isometric view of the detent ring of the access cover assembly of FIG. **2**;
(15) FIG. **14** shows a partially exploded isometric view of an alternative access cover assembly of the present disclosure;
(16) FIG. **15** shows an exploded isometric view of the access cover assembly of FIG. **14**; and
(17) FIG. **16** shows a partially exploded isometric cross-sectional view of still another alternative access cover assembly of the present disclosure.
(18) Corresponding references indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(19) Example embodiments will now be described more fully with reference to the accompanying drawings. The example embodiments are provided so that this disclosure will be thorough and fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.
(20) The present disclosure provides an access cover assembly **10** (e.g., FIGS. **2-4**) for machines, equipment, mechanical assemblies and like apparatus. The access cover assembly is configured to be mounted or attached to an exterior housing or casing of an apparatus (e.g., a mechanical assembly like a gear box) to cover an access port or opening in the housing that is used for gaining access to the interior of the apparatus.
(21) The access cover assembly **10** generally includes a cover plate **12** (e.g., FIGS. **8-10**) and a removable, threaded, self-locking plug **14** (e.g., FIGS. **11-12**) configured to detachably connect to the cover plate **12**. The cover plate **12** has an access opening **16** and the self-locking plug **14** can be secured to the cover plate **12**, e.g., via a threaded shaft **36** on the plug **14** that engages a threaded access opening **16** in the cover plate **12**. Further, the self-locking plug **14** incorporates a lock ring **20** that is co-operable with a detent interface **22** associated with the cover plate **12** to inhibit unintended rotation and loosening of the self-locking plug **14** when assembled to the cover plate **12**. In this regard, the detent interface **22** can be integrally formed in the cover plate **12** or alternatively be provided by way a separate detent ring **22***a* (e.g., FIG. **13**) that can be assembled with or securely affixed to the cover plate **12**. The self-locking plug **14** nevertheless can be quickly and easily installed and removed with a tool (e.g., a wrench or ratchet drive, see FIG. **5**, socket **33**) with routine effort and without possessing a special skill set.

(22) The removable, self-locking plug **14** eliminates the need to remove the entire cover plate **12** to gain access to the interior of the apparatus, particularly in circumstances where a smaller diameter port (i.e., relative to the diameter or size of the cover plate **12**) may satisfy the access to the apparatus that is required for conducting regular inspection or maintenance tasks. As such, it is unnecessary to cut and remove any lock wire or other fastener locking systems and all of the mounting bolts that secure the access cover to the apparatus, and then reattach the access cover and lockwire the mounting bolts when the task is complete.
(23) The access cover assembly **10** can include a cover plate **12** and a self-locking plug **14** removably attached to the cover plate **12**. As shown in FIGS. **9-11**, the cover plate **12** can have a central portion **24** and a mounting flange **26** disposed about a perimeter of the central portion **24**. The cover plate **12** can be appropriately shaped and sized to fit an opening, e.g., in the housing, of the apparatus. As such, the cover plate **12** can be designed in any of a variety of geometric configurations (e.g., polygonal, circular, hemispherical, etc.). The mounting flange **26** can be adapted to interface with the apparatus at the access opening **16** and can include a plurality of apertures **28** configured to accommodate a plurality of fasteners (e.g., mounting bolts) to secure the access cover assembly **10** to the apparatus.
(24) As shown in FIGS. **9** and **10**, the cover plate **12** can include a first aperture (or access opening) **16***a* having a first diameter d**1** and a second aperture **16***b* having a second diameter d**2** that is greater than the first diameter d**1**. The second aperture **16***b* is concentric with the first aperture **16***a*. The first aperture **16***a* and second aperture **16***b* can be located on a axis X of the access cover assembly **10** that is central to the cover plate **12** (e.g., FIG. **8**) or, alternatively, on an axis that is offset from the central axis X of the cover plate **12**. The first aperture **16***a* extends through the central portion **24** of the cover plate **12**. The first aperture **16***a* also defines a first inner circumferential wall **16***c* comprising an internal screw thread. The concentric, second aperture **16***b* extends into the central portion **24** for a depth that is less than a thickness of the central portion **24**. As such, the second aperture **16***b* is akin to a counter-bore about the first aperture **16***a*. The second aperture **16***b* comprises the detent interface **22**. In this respect, the second aperture **16***b* has a second inner circumferential wall **16***d*. In one aspect of the present disclosure, the second inner circumferential wall **16***d* comprises a cylindrical surface including a plurality of spaced-apart, concave-shaped detents **22***b* equally spaced about a perimeter of the second inner circumferential wall **16***b*. Adjacent detents **22***b* are separated from one another by discrete sections of the cylindrical surface of the inner circumferential wall **16***d* (best understood with reference to FIG. **13**). As seen in FIG. **13**, the number of detents shown is 12, although the number of detents can vary, with an exemplary range of between 8 and 16 detents.
(25) The self-locking plug **14** of the access cover assembly **10** is oriented along the axis X of and disposed in the first aperture **16***a* and the second aperture **16***b* of the cover plate **12**.
(26) As shown in FIGS. **12-13**, the self-locking plug **14** includes a main body **30** and a lock ring **32** that is attached to the main body **30**. The main body **30** comprises a head **34** and a threaded shaft **36**. The head **34** is configured to receive a tool (e.g., a wrench or ratchet drive) for rotating the self-locking plug **14** about a centerline or central axis X**1** of the self-locking plug **14** and the threaded shaft **36** includes an external screw thread that mates with (i.e., engages) the internal screw thread **16***c* of the access opening **16***a*.
(27) The lock ring **32** includes a cylindrical portion **38** that is attached to the head **34** of the main body **30** and a plurality of resilient, flexible (e.g., spring-like) beam fingers **40** that are arranged about a circumference of the cylindrical portion **38**. Each beam finger **40** extends from a proximal end **42** (at the cylindrical portion of the lock ring) to a distal end **44** that forms a fingertip **46**. Each fingertip **46** of the beam fingers **40** includes a plurality of beveled surfaces **48**. In an aspect of the disclosure, the beveled surfaces **48** of the fingertips **46** of the beam fingers **40** are located on the outer side of the beam fingers **40** (i.e., outwardly facing away from the centerline of the self-locking plug **14**). The fingertip **46** is configured to mechanically interact with (e.g., exhibit a bias

against) the detent interface **22** of the second aperture **16***b* so as to inhibit rotation of the self-locking plug **14** about the central axis X**1**. The number of beam fingers can vary, with a range of between about 12 to 20 being exemplary.

(28) In another aspect of the disclosure, as shown in FIG. **13**, a detent ring **22***a* that is separate from the cover plate **12** can provide the feature of the second inner circumferential wall **16***d* of the second aperture **16***b*. In this respect, the separate detent ring **22***a* can include the inner circumferential and cylindrical wall **22***c* comprising the plurality of detents **22***b*, and have an outer circumferential wall **22***d* comprising an outer cylindrical surface. The detent ring can then be secured (e.g., by an interference fit, welding, or otherwise) in the second aperture **16***b* of the cover plate **12**.

(29) In still another aspect of the disclosure, an alternate access cover assembly **100** is shown in FIGS. **15** and **16**. Like the access cover assembly **10** already discussed, the access cover assembly **100** shown in FIG. **14** generally includes a cover plate **120** and a removable, threaded, self-locking plug **140** configured to detachably connect to the cover plate **120**. The self-locking plug **140** incorporates a lock ring **320** that is co-operable with a detent interface **220** associated with the cover plate **120** to inhibit unintended rotation and loosening of the self-locking plug **140** when assembled to the cover plate **120**. The self-locking plug **140** nevertheless can be quickly and easily installed and removed with a tool.

(30) The cover plate **120** incorporates a detent interface **220** that is co-operable with the lock ring **320** to inhibit unintended loosening. In this respect, the cover plate **120** of the access cover assembly **100** of FIG. **14** can include a boss **220***a* projecting from the central portion **240**. The boss **220***a* can have a second diameter and include a second outer circumferential wall **220***d*. The outer circumferential wall **220***d* of the boss **220***a* can comprise a cylindrical surface in which are included a plurality of spaced-apart, concave-shaped detents **220***b* that are equally spaced about a perimeter of the second outer circumferential wall **220***d*. Adjacent detents **220***a* can be separated from one another by discrete sections of the cylindrical surface of the outer circumferential wall **220***d*. The number of detents can vary, and a range of between about 8 and 16 detents is exemplary.

(31) A first aperture **160***a*, having a first diameter and being concentric with the boss **220***a*, can extend through both the boss **220***a* and the central portion **240** of the cover plate **120**. The first aperture **160***a* can also include an inner circumferential wall **160***c* defining an internal screw thread. The self-locking plug **140** can engage the threaded access opening **160***c* when the self-locking plug **140** is secured to the cover plate **120** as further described herein.

(32) The boss **220***a* can define an interface port for the self-locking plug **140** and the first aperture **160***a* can define an access opening **160** through the cover plate **120**. As can be understood with reference to FIGS. **15** and **16**, the boss **220***a* can, alternatively, be integrally formed with the cover plate **120** or be a separate component that is securely affixed to the cover plate **120**. Further, it is also contemplated that the detent interface **220** can be integrally formed in the outer circumferential wall of the boss **220***a* or, alternatively, can be provided by way of a separate detent interface component (e.g., a detent ring) that can be assembled with or secured to the outer cylindrical surface **220***d* of the boss **220***a*.

(33) Also shown in FIGS. **15** and **16** is a removable, threaded, self-locking plug **140** configured to detachably connect to the cover plate **120**. The self-locking plug **140** includes a main body **300** and a lock ring **320** that is attached to the main body **300**. The main body **300** comprises a head **340** (e.g., a hexagonal head) and a threaded shaft **360**. The self-locking plug **140** is configured to accommodate a tool (e.g., a wrench or ratchet drive) for rotating the self-locking plug **140** about a centerline or central axis X**100** of the self-locking plug **140**. The threaded shaft **360** includes an external screw thread that engages the internal screw thread **160***c* of the access opening **160***a* of the cover plate **120**.

(34) The lock ring **320** of the self-locking plug **140** includes an upper section **322** that is attached to the head **340** of the main body **300** and a plurality of resilient, flexible (e.g., spring-like) beam

fingers **400** that are arranged about a circumference of the upper section **322**. Each beam finger **400** extends from a proximal end **420** (at a cylindrical portion of the upper section **322** of the lock ring **320**) to a distal end **440** that forms a fingertip **460**. Each fingertip **460** of the beam fingers **460** includes a plurality of beveled surfaces **480**. In an aspect of the disclosure, the beveled surfaces **480** of the fingertips **460** of the beam fingers **400** are located on an inner side of the beam fingers **400** (i.e., inwardly facing the centerline X**100** of the self-locking plug **140**). The fingertip **460** is configured to mechanically interact with (e.g., exhibit a bias against) the detent interface **220** (e.g., on the boss **220***a*) of the cover plate **120** so as to inhibit rotation of the self-locking plug **140** about the central axis X**100**. The number of beam fingers can vary, with a range of between about 12 to 20 being exemplary.

(35) As can be understood with reference to FIG. **15**, it is contemplated that the upper portion **322** of the lock ring **320** can include an aperture **324** into which the head **340** of the main body **300** can be disposed. As shown in the FIGS. **15-16**, the aperture **324** and head **340** are hexagonally shaped. It is further contemplated that a fixed connection between the aperture **324** of the lock ring **320** and the head **340** of the main body **300** can be achieved by any of a variety of known connection methods, including an interference fit, welding or the like. Once assembled, the head portion **340** of the self-locking plug **140** provides a ready interface for a tool (e.g., a wrench or ratchet drive).

(36) Yet another alternative embodiment of an access cover assembly **1000** of the present disclosure is shown in FIG. **16**. In the embodiment of the access cover assembly **1000** shown in FIG. **16**, like the access cover assemblies **10**, **100** discussed above, the access cover assembly **1000** shown in FIG. **16** generally includes a cover plate **1200** and a removable, threaded, self-locking plug **1400** configured to detachably connect to the cover plate **1200**. The self-locking plug **1400** incorporates a lock ring **3200** that is co-operable with a detent interface **2200** associated with the cover plate **1400** to inhibit unintended rotation and loosening of the self-locking plug **1400** when assembled to the cover plate **1200**. The self-locking plug **1400** nevertheless can be quickly and easily installed and removed with a tool.

(37) The cover plate **1200** incorporates a detent interface **2200** that is co-operable with the lock ring **3200** to inhibit unintended loosening. In this respect, the cover plate **1200** of the access cover assembly **1000** of FIG. **16** can include a boss **2200***a* projecting from the central portion **2400**. The boss **2200***a* can have an outer diameter and include an outer cylindrical wall **2200***d*. The outer cylindrical wall **2200***d* of the boss **2200***a* can define an external screw thread. The self-locking plug **1400** can engage the threaded outer cylindrical wall **2200***d* of the boss **2200***a* when the self-locking plug **1400** is secured to the cover plate **1200**, as further described herein.

(38) A first aperture **1600***a*, having a first diameter can extend through both the boss **2200***a* and the central portion **2400** of the cover plate **1200**. A second aperture **1600***b* in the boss **2200***a* having a second diameter that is greater than the first diameter and being concentric with the outer cylindrical wall **2200***d* of the boss **2200***a*, has a depth that is less than a height of the boss **2200***a*. The second aperture **1600***b* can include the detent interface **2200** having a plurality of spaced-apart, concave-shaped detents **2200***b* that are equally spaced about a perimeter of the second aperture **1600***b*. Adjacent detents **2200***b* can be separated from one another by discrete cylindrically shaped sections of the second aperture **1600***b*. The number of detents can vary with a range of between about 8 and 16 detents being exemplary.

(39) The boss **2200***a* can define an interface port for the self-locking plug **1400** and the first aperture **1600***a* can define an access opening **1600** through the cover plate **1200**. Like the previously described access cover assemblies **10**, **100**, and as can be understood with reference to FIG. **16**, the boss **2200***a* can, alternatively, be integrally formed with the cover plate **1200** or be a separate component that is securely affixed to the cover plate **1200**. Further, it is also contemplated that the detent interface **2200** can be integrally formed in the second aperture **1600***b* of the boss **2200***a* or, alternatively, can be provided by way of a separate detent interface component (e.g., a detent ring **22***a*, FIG. **13**) that can be secured within the second aperture **1600***b* of the boss **2200***a*.

(40) Also shown in FIG. **16** is a removable, threaded, self-locking plug **1400** configured to detachably connect to the cover plate **1200**. The self-locking plug **140** can comprise a cylindrical, cap-like main body **3000** and a lock ring **3200** that is attached to the main body **3000**. The main body **3000** can include a sleeve **3002** comprising an internal, cylindrically shaped surface **3004** defining an internal screw thread. It can be understood that the self-locking plug **1400** can be configured to accommodate a tool (e.g., a wrench or ratchet drive) for rotating the self-locking plug **1400** about a centerline or central axis X**1000** of the self-locking plug **1400** (e.g., an exterior of the main body **3000** can incorporate tool-engaging "flats", such as for a wrench or socket). The internally threaded **3004** sleeve **3002** can engage the external screw thread **2200***c* of the boss **2200***a* of the cover plate **1200** when the self-locking plug **1400** is connected to the cover plate **1200**.
(41) The lock ring **3200** of the self-locking plug **1400** includes an upper section **3220** that is attached to the main body **3000** and a plurality of resilient, flexible (e.g., spring-like) beam fingers **4000** that are arranged about a circumference of the upper section **3220**. Each beam finger **4000** extends from a proximal end **4200** (at a cylindrical portion of the upper section **3220** of the lock ring **3200**) to a distal end **4400** that forms a fingertip **4600**. Each fingertip **4600** of the beam fingers **4000** includes a plurality of beveled surfaces **4800**. The number of beam fingers can vary, with a range of between about 12 to 20 being exemplary.
(42) In an aspect of the access cover assembly **1000** shown in FIG. **16**, the beveled surfaces **4800** of the fingertips **4600** of the beam fingers **4000** can be located on the outer side of the beam fingers **4000** (i.e., outwardly facing away from the centerline X**1000** of the self-locking plug **1400**). The fingertip **4600** is, thus, configured to mechanically interact with (e.g., exhibit a bias against) the detent interface **2200** of the second aperture **1600***b* so as to inhibit rotation of the self-locking plug **1400** about the central axis X**1000**.
(43) In still another aspect of the disclosure, the access cover assembly **10**, **100**, **1000** may incorporate one or more seals (see, e.g., FIG. **6**) to provide fluid-tight interfaces between the access cover assembly **10**, **100**, **1000** and the apparatus. For example, a seal or gasket **50** may be employed between the cover plate **12**, **120**, **1200** (e.g., at the mounting flange **26**, **260**, **2600**) and the mounting surface of the apparatus. Alternatively, or in addition, a seal such as an O-ring **52**, can be included between the self-locking plug **14**, **140**, **1400** and the cover plate **12**, **120**, **1200** (e.g., between a threaded shaft **36**, **360** or sleeve **3002** and a cover plate aperture **16***a*, **160***a* or boss **2200***d*).
(44) In another aspect of the invention, the dimension(s) of the access opening **16**, **160**, **1600** can be varied to accommodate different size(s) (e.g., within the overall dimensions of the cover plate **12**, **120**, **1200**) to provide any needed or desired access to the apparatus. Additionally, the size of the self-locking plug **14**, **140**, **1400** can likewise be varied to correspond to the various size(s) of the access opening **16**, **160**, **1600**.
(45) The access cover assembly **10**, **100**, **1000** of the present disclosure can be retro-fit to machines, equipment, mechanical assemblies and like apparatus in the field, replacing existing access covers. Still further, existing access covers can be reworked, re-purposed or reconfigured to the access cover assembly of the present disclosure.
(46) The access cover assembly **10**, **100**, **1000** of the present disclosure enables maintenance and/or inspection tasks to be performed more efficiently, reducing apparatus downtime and labor.
(47) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. An access cover assembly comprising: a cover plate and a self-locking plug removably attached to the cover plate; the cover plate comprising a central portion and a mounting flange positioned about a perimeter of the central portion; the flange comprising a plurality of apertures configured to accommodate a plurality of mounting bolts; the cover plate comprising a first aperture having a first diameter and a second aperture having a second diameter that is greater than the first diameter; the first aperture being centered on a central axis and extending through the central portion, the first aperture comprising a first inner circumferential wall comprising an internal screw thread; the second aperture being centered on the central axis and concentric with the first aperture, the second aperture extending into the central portion to a depth that is less than a thickness of the central portion; a second inner circumferential wall defining one of the first and second apertures and presenting a plurality of concave-shaped detents equally spaced about a perimeter of the second inner circumferential wall, wherein adjacent detents are separated from one another by a generally planar surface; the self-locking plug being disposed in the first aperture and the second aperture along the central axis; the self-locking plug comprising a main body and a lock ring, coaxially aligned with the central axis; and the lock ring attached to the main body; the main body comprising a head and a shaft, the head being configured to receive a tool for rotating the self-locking plug about the central axis, the shaft comprising an external screw thread configured to engage the internal screw thread of the first aperture of the cover plate in a first rotational direction to couple the self-locking plug to the cover plate; the lock ring comprising a cylindrical portion attached to the head and a plurality of resilient, flexible beam fingers arranged about a circumference of the main body, each beam finger extending from a proximal end at the cylindrical portion to a distal end comprising a fingertip spaced from the main body; each fingertip of the plurality of beam fingers comprising an outer surface that presents beveled surfaces and a substantially planar surface between the beveled surfaces, with the beveled surfaces being beveled radially inward relative to the central axis; the plurality of beam fingers being normally biased radially outwardly relative to the central axis and against the second inner circumferential wall of the second aperture; and wherein the radially outwardly bias of the plurality of beam fingers causes the beveled surface of each respective fingertip to mechanically interact with the detents of the second inner circumferential wall of the second aperture to inhibit unintended rotation and loosening of the self-locking plug about the central axis.
2. The access cover assembly of claim 1, wherein the second inner circumferential wall is presented on an inner detent ring surface of a detent ring fixedly disposed in the second aperture, wherein the inner detent ring surface is the second inner circumferential wall and presents the plurality of concave-shaped detents.
3. The access cover assembly of claim 2, wherein the detent ring presents an inner surface and an outer surface, and wherein the inner surface comprises the plurality of concave-shaped detents that are equally spaced about the perimeter of the second inner circumferential wall, and the outer surface is configured to couple the detent ring to the second aperture.
4. The access cover assembly of claim 1, wherein the bias of the beam fingers results from a resiliency and flexibility of the beam fingers.